COLX 581 Report - Auditing LLM Summaries

Jai Sharma[1], **Gilly Guo**[1], **Rayan Jaz**[1] and **Sina Oloomi**[1]

[1]University of British Columbia

1 Abstract

As Large Language Models are increasingly integrated into professional and academic workflows for automated summarization, their reliability on these tasks remains a critical concern, particularly in nuanced linguistic domains. This project investigates the reliability of LLM-generated summaries within the context of argumentative discourse, utilizing a corpus derived from the Reddit subreddit r/changemyview. We evaluate the capacity of Gemini 3.0 Fast to accurately capture multi-party debates, identify shifts in participant perspectives, and maintain factual integrity across complex rhetorical structures. We then manually annotated these summaries for accuracy, brevity, and sentiment, and trained machine learning models to approximate the human accuracy judgment. We progressed from simple TF-IDF baselines to Multi-Task Learning architectures and data-centric augmentation strategies. Our findings demonstrate that while base models struggle with class imbalance, the introduction of an auxiliary brevity task improves minority class recovery. Our final confidence-gated hybrid model achieved a peak accuracy of 0.750 and a Macro F1 of 0.415, while Sprint 4 data augmentation showed that carefully selected extra data is more useful than simply adding larger noisy pools.

2 Introduction

The rapid proliferation of Large Language Models has transformed document processing, with automated summarization serving as a primary use case for increasing workplace efficiency. However, the "black box" nature of these models introduces significant risks when users over-rely on summaries without verifying the underlying source material. This risk is amplified in argumentative contexts where the summary must not only compress information but also preserve the logical flow, identify conflicting stances, and detect potential shifts in opinions as the argument progresses. While existing evaluation metrics often focus on lexical overlap or fluency, they frequently fail to capture the semantic depth and "reasoning fidelity" required for high-stakes decision-making.

Our study addresses this gap by focusing on threads from the r/changemyview community, a subreddit where an Original Poster (OP) posts an opinion that is then challenged by various responders, potentially leading to the OP changing their mind. This domain presents a novel challenge for AI; it requires a high degree of social reasoning to establish the exact arguments occurring between the OP and their various challengers, and to pinpoint the exact moment of cognitive change, or the lack thereof. In this report, we present the full modeling pipeline developed across COLX 581: a reasoned train/dev/test split, two baseline classifiers, ensembling and transfer learning, a Multi-Task Learning objective, and data augmentation through bootstrapping, active learning, and few-shot synthetic examples. We argue that assessing AI summarization abilities through the lens of argumentative text is essential for building more robust, trustworthy systems that can navigate the complexities of human language.

3 Data

3.1 Corpus Acquisition and Composition

Our dataset consists of 400 annotated r/changemyview (r/CMV) argument branches, each paired with a summary generated by Gemini 3.0 Fast. The initial corpus was built from r/CMV data accessed through ConvoKit (Chang et al., 2020), with additional r/CMV material collected through our scraper. A further 100 annotated examples were later reserved for Sprint 4 data augmentation. All branches were at least 5 comments long, including the original post, and

Gemini 3.0 Fast was prompted to produce a summary under 200 words. A unique feature of this corpus is its paired structure: for most top-level original posts, we include two branched threads—one where the OP acknowledges their viewpoint was changed to some degree, and one where it was not.

3.2 Annotation Schema

To evaluate the Gemini 3.0 Fast summaries, four human annotators rated each output across three criteria:

- **Sentiment (Success):** A boolean label (0 or 1) indicating if the model correctly identified whether the OP's mind was changed.
- **Accuracy:** A 0–2 scale (0 = not accurate, 1 = moderately accurate, 2 = very accurate) assessing how well the model represented the core arguments.
- **Brevity:** A 0–2 scale (0 = few points, 1 = some points, 2 = most points) evaluating the model's ability to capture the essential "key points" of the thread.

Figure 1 shows the distribution of scores given by the human annotators.

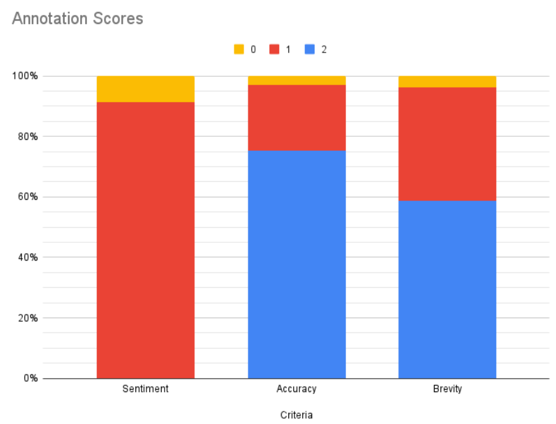

Figure 1: Overall scores given by human annotators.

To ensure scientific validity, we calculated inter-annotator agreement using Krippendorff's Alpha (Krippendorff, 2011). This metric was chosen for its ability to handle multiple annotators and its support for ordinal data, which was necessary for our 0–2 scales, where a disagreement between 0 and 2 is more significant than one between 1 and 2.

3.3 Reasoned Data Split

To prepare for baseline model training, we implemented a "reasoned split" rather than a random 80-10-10 shuffle. To prevent data leakage, we split the data by thread number rather than by row. This ensures that the successful and unsuccessful branches of the same discussion remain in the same set, preventing the model from seeing the shared context of a test thread during training.

Furthermore, we stratified the split by reviewer id to maintain a balanced mix of human perspectives across sets. The final distribution resulted in 320 rows for training, 40 for development, and 40 for testing. While the dataset exhibits a natural class imbalance—most summaries were scored relatively highly on all criteria by the human annotators—this distribution was maintained across splits to reflect a realistic evaluation environment.

4 Methodology

The primary objective of this project phase is to establish a robust baseline for predicting the human-assigned Accuracy score (0, 1, or 2) of the summaries generated by Gemini 3.0 Fast. By treating this as a supervised classification task, we aim to determine the extent to which traditional and simple neural architectures can mirror human judgment of "reasoning fidelity" in argumentative discourse.

4.1 Experimental Setup and Data Leakage Prevention

All experiments used the thread-aware train/dev/test split described in §3.3, which reduced leakage across paired branches from the same discussion and preserved reviewer balance across splits.

4.2 Feature Representation

For each example, we concatenated the original source argument and the AI-generated summary using "SOURCE:" and "SUMMARY:" markers. Text was represented with TF-IDF unigram and bigram features, English stop-word removal, a minimum document frequency of 2, and a maximum vocabulary of 5,000 features.

4.3 Baseline Models

We compared a balanced Logistic Regression classifier against a simple Multi-Layer Perceptron (MLP). Logistic Regression provided an interpretable small-data baseline, while the MLP tested whether a nonlinear classifier over SVD-reduced TF-IDF features could improve performance. Both models were implemented with scikit-learn (Pedregosa et al., 2011) and evaluated with Macro F1 as the primary metric because it better reflects performance across imbalanced classes than raw accuracy.

4.4 Ensemble Methods

We combined the traditional and neural models with a weighted soft-voting ensemble. Because Logistic Regression performed better in early trials, a manual weight search assigned it 75% of the ensemble weight and the MLP 25%.

4.5 Transfer Learning

To supplement sparse TF-IDF features, we integrated pretrained GloVe word embeddings (glove-wiki-gigaword-50) (Pennington et al., 2014). We averaged embeddings separately for the source argument and generated summary, then concatenated them with the baseline feature representations. This added an external semantic signal that could capture lexical relationships unavailable to TF-IDF alone.

4.6 Multi-Task Learning

The third iteration of the project shifted focus from input feature engineering to output regularization through the implementation of Multi-Task Learning (MTL). Rather than modifying the input space, the architecture was redesigned to jointly predict the primary Accuracy label alongside a secondary Brevity objective. Brevity was selected as the auxiliary task because it demonstrated a more balanced class distribution than sentiment and shared a high semantic correlation with human-judged summary accuracy. A Traditional MTL model, utilizing Logistic Regression on a combined joint-label space, achieved the best pure Macro F1 of 0.421. This approach effectively mitigated severe class imbalance by recovering recall on the minority "moderately accurate" class (Class 1), raising its F1 score from 0.286 to 0.455.

4.7 Data-Centric Augmentation

Building on the optimized MTL architecture, Sprint 4 tested whether targeted data expansion could improve the low-resource r/CMV classifier:

Bootstrapping: The tuned Sprint 3 traditional MTL model pseudo-labeled 100 additional r/CMV threads, equivalent to 25% of the original dataset. We compared confidence cutoffs to avoid adding low-confidence silver labels.

Active Learning: Examples were ranked with entropy-based uncertainty sampling, and we retrained after adding human labels for the top 5%, 10%, 15%, and 20% most uncertain examples.

Few-shot Synthetic Expansion: We expanded training data by 20% with GPT 5.4-generated r/CMV-style examples, then compared the original synthetic set against a balanced-by-accuracy resampling.

Ablations compared random selection against uncertainty ranking and tested the relative importance of the MTL objective versus GloVe transfer features.

5 Results and Analysis

Performance improved iteratively throughout the sprints (see Table 1). The largest gain came from combining low-resource modeling choices that matched the structure of the annotation task: class-balanced traditional classification, GloVe transfer features, and finally an MTL objective that linked Accuracy with Brevity. The Traditional MTL model improved minority class recovery, raising the Class 1 F1-score from 0.286 to 0.455, while the confidence-gated hybrid achieved the highest raw accuracy.

Model	Acc.	Macro F1
S1 Traditional Baseline	0.650	0.336
S2 Weighted Ensemble	0.725	0.372
Traditional MTL (LogReg)	0.700	0.421
Confidence-gated (S2+S3)	0.750	0.415

Table 1: Performance comparison across sprints on the development set.

Sprint 4 did not surpass the best Sprint 3 hybrid, but it did clarify which augmentation strategies were useful under a fixed low-resource budget. Table 2 shows that entropy-based active learning was the strongest Sprint 4 strategy: adding the top 5% or 15% uncertain human-labeled examples matched the Sprint 3 Traditional MTL score of 0.700 accuracy and 0.421 Macro F1. In contrast, adding the full pseudo-labeled pool reduced performance, showing that more data was not automatically better when the added labels were noisy. Few-shot synthetic examples were also difficult to use effectively: the original synthetic set decreased performance substantially, while balancing by Accuracy improved accuracy but still produced a weaker Macro F1 than the active-learning condition.

Sprint 4 Condition	Acc.	Macro F1
Active learning, top 5% entropy	0.700	0.421
Active learning, top 15% entropy	0.700	0.421
Bootstrapping, best cutoff	0.675	0.407
Bootstrapping, full pool	0.625	0.380
Few-shot synthetic, original	0.575	0.305
Few-shot synthetic, balanced	0.675	0.348

Table 2: Sprint 4 augmentation results on the development set.

The ablation tests support this interpretation. At the same 15% budget, entropy selection reached 0.700 accuracy and 0.421 Macro F1, while random selection only reached 0.625 accuracy and 0.380 Macro F1. This suggests that the active-learning ranking identified examples that were genuinely informative for the model rather than simply increasing the training set size. Removing the MTL objective preserved accuracy at 0.700 but reduced Macro F1 to 0.384, and removing GloVe transfer reduced Macro F1 further to 0.360. These ablations indicate that the final model's useful behavior came from both output-side regularization and external semantic features, not from a single component.

5.1 Qualitative Analysis

Comparing the S1 baseline to the S3 MTL model revealed that the MTL objective acts as a useful regularizer. The early models often over-predicted Class 2 because most summaries in the corpus were rated as highly accurate. In fixed cases, the MTL model moved predictions from Class 2 down to Class 1, correctly identifying that summaries could be broadly faithful while still missing important argumentative details. However, some "broken" cases occurred where the model's increased caution led to under-prediction on strong summaries. Notably, Class 0 remained undetected across all models due to its extreme rarity in the corpus, suggesting a hard ceiling for lexical and averaged-embedding features when the task requires detecting severe factual failure or argumentative distortion.

The prevalence of Class 2 labels should also be interpreted as an outcome of the summarization task itself, not only as a weakness of the dataset. Gemini 3.0 Fast often did well at capturing the main argumentative exchange: the OP's position, the responder's challenge, and whether the discussion produced a meaningful shift. However, since the summaries were constrained to under 200 words, annotators primarily judged whether the central argument was preserved rather than whether every local detail was included. Minor omissions were therefore difficult to penalize when they did not change the overall argument, and artificially lowering scores to balance the dataset would have made the labels less faithful to the annotation guidelines. This suggests an important future reframing: the same model might receive lower scores in a setting where small factual details, evidence chains, or quotation-level precision are essential, but r/CMV argument summaries reward accurate high-level compression.

The most persistent failure was Class 0. There were very few truly inaccurate summaries in the annotated data, meaning our models had almost no positive evidence for what a failed argumentative summary looked like. This imbalance likely emerged because Gemini 3.0 Fast was generally good at the specific task we gave it: produce a short, high-level account of an argument. The summaries did not need to preserve every supporting detail to receive a high score; they mainly needed to capture the OP's stance, the responder's challenge, and the overall argumentative outcome. Under that rubric, minor omissions were often not serious enough to justify a Class 0 label, and assigning lower scores just to balance the dataset would have made the labels less valid. As a result, Class 0 remained undetected across models, which placed a ceiling on Macro F1 even when overall accuracy looked reasonable. This is not simply a modeling error: if the training data mostly contains acceptable summaries, a supervised classifier will predominantly learn the difference between "moderately accurate" and "very accurate" much more reliably than the boundary between "somewhat flawed" and "not accurate." Future versions of the task would need either more naturally occurring bad summaries or a deliberately constructed challenge set of hallucinated, stance-flipped, detail-sensitive, or argument-missing summaries to evaluate severe failure detection.

Sprint 4 also reiterates a lot of these points, that is, active learning helped because the highest entropy examples all followed a similar pattern where they were on the border between "moderately accurate" and "very accurate," where additional labels gave the model better evidence about borderline judgments. By contrast, pseudo-labeling and synthetic examples were more likely to amplify the model's existing bias toward common labels. That is why the best Sprint 4 result was not a larger model or a larger training set, but a more selective data strategy.

6 Conclusion

This project demonstrates that auditing LLM summaries in argumentative domains requires more than fluency-based evaluation. Our strongest result came from a confidence-gated hybrid model with 0.750 accuracy and 0.415 Macro F1, while the Traditional MTL model produced the best pure Macro F1 at 0.421 by improving recovery of the minority "moderately accurate" class. Sprint 4 showed that data-centric methods are valuable, but only when the added data is carefully selected: entropy-based active learning matched the Sprint 3 MTL result and outperformed random selection, while noisy pseudo-labels and synthetic examples generally weakened the classifier. Overall, the project supports a practical conclusion for low-resource NLP: when labels are scarce and subjective, the structure of the label space and the quality of added examples matter more than simply increasing dataset size.

Limitations

The primary limitation was our small dataset size, which made our models susceptible to noise and limited the absolute gains possible via supervised learning. Furthermore, the Accuracy label remains subjective; while inter-annotator agreement was established, human judgment of "persuasion" is inherently variable. The task definition also produced a naturally high-scoring label distribution because concise argument summaries can be accurate even when they omit lower-level details. Finally, the reliance on surface-level lexical and embedding features limited our models' abilities to detect errors and logical fallacies that did not manifest as simple keyword mismatches.

Ethical Considerations

The use of Reddit data from r/changemyview carries risks related to speaker bias and the potential for offensive content in controversial threads. We mitigated this by ensuring the dataset was used for evaluative auditing purposes only. Additionally, the development of models that "judge" the accuracy of persuasion must be handled with care to avoid creating systems that could be repurposed for automated manipulation or the suppression of dissenting viewpoints. All training was conducted on publicly available data with metadata anonymization where possible.

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